Problem–01

Title

Next Prime After N

Topics

Number Theory, Prime Numbers, Conditional Statements, Loops, Nested Loop (Optimization optional)

Difficulty Level

Medium

Problem Detailed Statement (Scenario Based)

In a competitive coding classroom at GLA University, students are learning about number theory and prime numbers. During a practice session, the trainer gives a challenge:

Every student is given a positive integer **N**, and their task is to find the **next immediate prime number greater than N**.

However, the problem is not as simple as it sounds. Students are not allowed to use any built-in prime checking libraries. They must:

- Use proper looping constructs.
- Apply conditional logic carefully.
- Optimize prime checking using mathematical understanding.
- Avoid unnecessary computations.

A prime number is defined as a number greater than 1 that has only two distinct divisors: 1 and itself.

For example:

- If N = 10, the next prime is 11.
- If N = 17, the next prime is 19.
- If N = 1, the next prime is 2.

The program must continue checking numbers greater than N until it finds the next prime number.

This problem tests:

- Understanding of prime logic
- Efficient loop control

- Break conditions
- Mathematical thinking
- Edge case handling ($N \leq 1$)

Your task is to write a program that reads an integer N and prints the next prime number strictly greater than N.

Input Format

- A single integer **N**
- Constraint:
- `0 ≤ N ≤ 10^6`

Output Format

- Print a single integer representing the next prime number greater than N.

Examples

Example 1

Input:

```
10
```

Output:

```
11
```

Example 2

Input:

```
17
```

Output:

```
19
```

Example 3

Input:

```
1
```

Output:

```
2
```

Example 4

Input:

```
24
```

Output:

```
29
```